# Arenula

MCP server suite for the s&box game engine

Get Started | View Source

| 19 | ~120 | 3 | 8-13K |
|---|---|---|---|
| TOOLS | ACTIONS | SERVERS | TOKENS |

## Three Servers, One Suite

Each server has a distinct role. The Editor runs inside s&box; the API and Docs servers run standalone.

### Editor

Scene manipulation, code compilation, asset management, terrain editing. Runs inside the s&box editor as a C# library plugin.

19 tools  SSE :8098  C#

### API

Offline type and member reference. Indexes ~1,800 types and ~15,000 members from the s&box CDN schema.

8 tools  stdio  TypeScript

### Docs

Narrative documentation fetcher. Scrapes docs.facepunch.com, converts to Markdown, caches with LRU.

3 tools  stdio  TypeScript

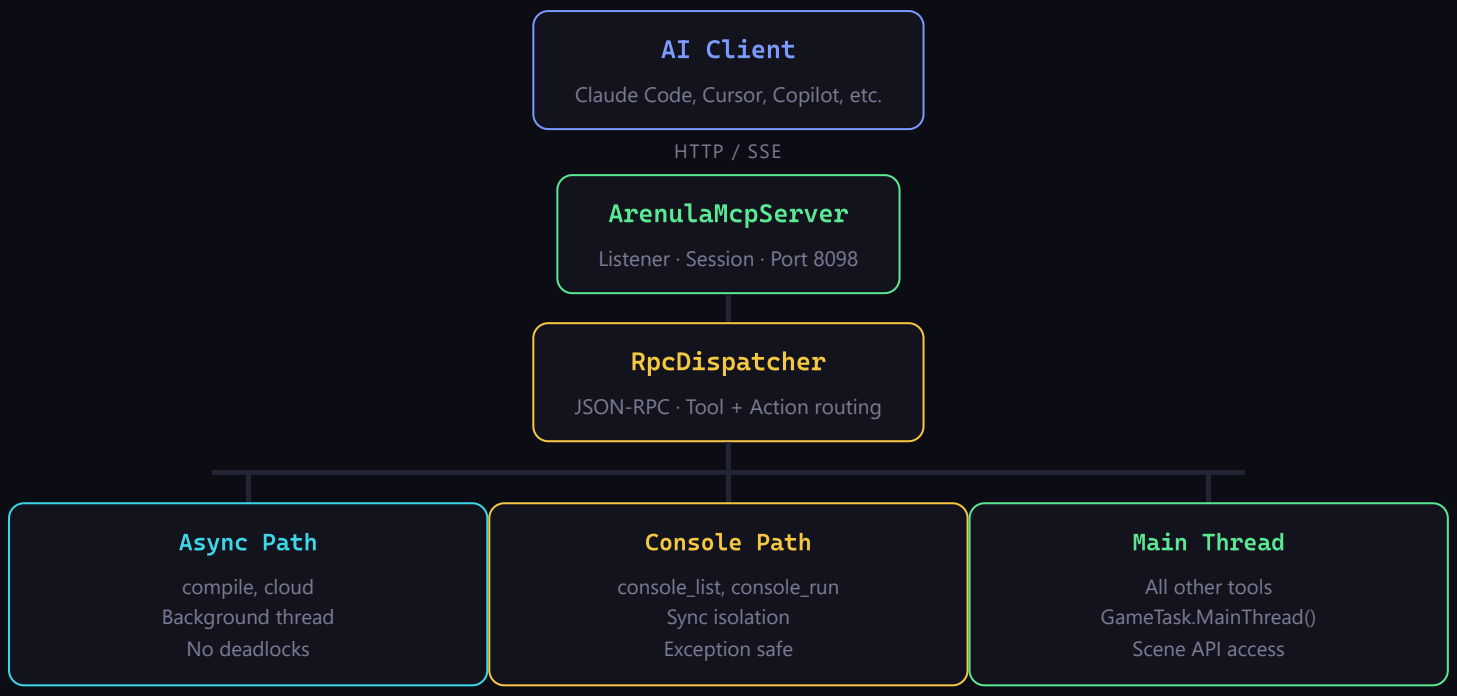
## Editor Tools

19 omnibus tools, ~155 actions. Each takes a required `action` parameter to select the operation.

| | | |
|---|---|---|
| **scene** | summary hierarchy statistics find find_in_radius get_details prefab_instances | 7 |
| **gameobject** | create destroy duplicate reparent rename enable set_tags set_transform batch_transform | 9 |
| **component** | add remove set_property set_enabled get_properties get_types copy | 7 |
| **compile** | trigger status errors generate_solution wait | 5 |
| **prefab** | instantiate get_structure get_instances break update create save_overrides revert get_overrides | 9 |
| **asset_query** | browse search open get_dependencies get_model_info get_material_properties get_mesh_info get_bounds get_unsaved get_status get_json get_references | 12 |
| **asset_manage** | create delete rename move save reload get_references | 7 |
| **editor** | select get_selected set_selected clear_selection frame_selection get_play_state start_play stop_play get_log save_scene save_scene_as undo redo console_list console_run open_code_file get_preferences set_preference | 18 |
| **session** | list set_active load_scene | 3 |
| **lighting** | create configure create_skybox set_skybox | 4 |
| **physics** | add_collider configure_collider add_rigidbody create_model_physics create_character_controller create_joint | 6 |
| **audio** | create configure | 2 |
| **effects** | create configure_particle configure_post_processing | 3 |
| **camera** | create configure capture_viewport | 3 |
| **mesh** | create_block create_plane create_cylinder create_wedge create_arch create_clutter extrude_faces remove_faces add_face clip_faces scale_mesh thicken_faces bevel_edges bevel_vertices split_edges quad_slice_faces dissolve_edges bridge_edges connect_vertices flip_faces extend_edges set_face_material set_texture_params set_vertex_position set_vertex_color set_vertex_blend get_info | 27 |
| **navmesh** | create_agent create_area create_link generate get_status query_path | 6 |
| **cloud** | search get_package mount | 3 |
| **project** | get_collision set_collision_rule get_input get_info | 4 |
| **terrain** | create configure get_info get_height get_height_region set_height noise erode stamp add_material remove_material get_material_at blend_materials set_hole paint_material import_heightmap export_heightmap sync | 18 |

## Architecture

28 C# source files organized into Core infrastructure and per-tool Handlers.




### Core 8 files

`ArenulaMcpServer` HTTP/SSE listener and session management. `RpcDispatcher` extracts tool name + action, routes to the correct handler. `ToolRegistry` holds all 19 tool schemas. `HandlerBase` provides response formatting, pagination, and error helpers. `SceneHelpers` handles scene resolution and tree walking.

### Handlers 20 files

One file per tool. Each exports a `Handle(action, args)` method with an internal switch on the action enum. Async handlers ( `CompileHandler` , `CloudHandler` ) use `HandleAsync` and dispatch before `GameTask.MainThread()` to prevent deadlocks.

### Threading Model

Scene APIs must run on the main thread. The dispatcher calls `await GameTask.MainThread()` before invoking sync handlers. Async handlers (network/compile) run on the background thread and may internally switch to main thread for specific sub-operations.

## Design Principles

Following Anthropic's MCP tool design guidance.

### Omnibus Tools

Related operations grouped with action enum parameter. Fewer tools means less selection ambiguity for the LLM.

### Rich Descriptions

3-4 sentences per tool explaining what it does, when to use it, and when not to. Negative guidance prevents misuse.

### Trimmed Responses

Canonical {id, name} tuples. Pagination with offset/limit. 16K character truncation. Concise and detailed modes.

### Actionable Errors

Structured errors with action context and suggestions. Fuzzy type matching: "Did you mean Rigidbody?"

### Schema Safety

additionalProperties: false on every schema. Enum constraints on actions and sub-types. Prevents hallucination.

### Async Safety

Network tools dispatch before MainThread to avoid deadlocks. Never block the main thread with synchronous waits.

## Installation

### 1. Copy the Editor plugin

```
# Copy into your s&box project's Libraries folder
Arenula-MCP/editor/  ⟶  YourProject/Libraries/arenula_mcp/
```

### 2. Build the API and Docs servers

```
# From the Arenula-MCP directory
cd api && npm install && npm run build
cd ../docs && npm install && npm run build
```

### 3. Configure .mcp.json in your project root

```
{
  "mcpServers": {
    "editor": { "type": "sse", "url": "http://localhost:8098/sse" },
    "api":    { "command": "node", "args": ["path/to/Arenula-MCP/api/dist/index.js"] },
    "docs":   { "command": "node", "args": ["path/to/Arenula-MCP/docs/dist/index.js"] }
  }
}
```

### 4. Open s&box

Arenula Editor compiles automatically and starts the MCP server on port 8098.

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Based on Ozmium MCP Server by ozmium7 · API from sbox-api-mcp by sofianebel · Docs from sbox-docs-mcp by sofianebel
GPL-3.0 (Editor) · MIT (API, Docs)